

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

End Semester Examination – Winter 2018

Course: B. Tech in Civil Engineering

Subject Name: Building Construction

Max Marks: 60

Date: 10/12/2018

Sem: III

Subject Code: BTCVC305

Duration: 3 Hr.

Instructions to the Students:

1. Solve **ANY FIVE** questions out of the following.
2. The level question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.
5. All sketches should be labeled, neat and proportionate

	(Level/CO)	Marks
Q. 1 Solve Any Two of the following.		
A) State the comparative merits and demerits of stone masonry and brick masonry?	(CO 1)	6
B) Draw neat sketches, showing plan of two consecutive courses of 1½ brick thick corner of wall in English bond?	(CO 1)	6
C) What are the general principles observed in brick masonry?	(CO 1)	6
Q.2 Solve Any Two of the following.		
A) What are the chief ingredients of plain cement concrete? Describe in detail the properties of each?	(CO 2)	6
B) Describe the properties of fresh and hardened concrete?	(CO 2)	6
C) Explain the role of admixtures in concrete?	(CO 2)	6
Q. 3 Solve Any One of the following.		
A) Sketch a semicircular arch showing all technical terms used in arch? Sketch the entire arrangement showing the details of centering work?	(CO 3)	12
B) Explain Brick lintel, reinforced concrete lintel and steel lintel with sketches?	(CO 3)	12
Q.4 Solve Any Two of the following.		
A) Draw a neat sketch of Frame and Panelled door? (plan , section & elevation)	(CO 3)	6
B) Draw a neat sketch of Louvered window?	(CO 3)	6
C) Draw a neat sketch of wooden stair showing all components?	(CO 3)	6
Q. 5 Solve Any One of the following.		
A) Explain king post truss and its joints with neat sketches?	(CO 3)	12
B) Explain Queen post truss and its joints with neat sketches?	(CO 3)	12
Q. 6 Solve Any two of the following.		
A) Explain merits and demerits of prefabrication?	(CO 4)	6
B) Describe various types of prefabrication? Explain the term ‘Tolerance ‘?	(CO 4)	6
C) Describe various guidelines for transportation and erection in prefabrication?	(CO 4)	6

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DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

End Semester Examination – MAY 2019

Course: B. Tech in Civil Engineering

Sem: III

Subject Name: Building Construction

Subject Code: BTCVC305

Max Marks: 60

Date: 01/06/2019

Duration: 3 Hr.

Instructions to the Students:

1. Solve **ANY FIVE** questions out of the following.
2. The level question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Solve of the following.		12
A) Draw the neat plan and elevation (not to scale) of the following bonds	Analysis	6
1. English bond - One Brick thick		2
2. English bond - One and half brick thick		2
3. English bond - two brick thick		2
B) Enlist the principles to be observed during construction of composite masonry?	Knowledge	6
Q.2 Solve of the following.		12
A) Enlist the type of admixtures with functions.	Knowledge	6
B) Describe the factors affecting to the strength of concrete?	Understand	6
Q. 3 Solve Any Two of the following.		12
A) With neat sketch describe the technical terms used in arches?	Application	6
B) I) Write down the difference between arches and lintels.	Analysis	4
II) Define the primary and the secondary beam.	Remember	2
C) I) What is the requirement of good formwork?	Knowledge	4
II) What are the functions of lintel?	Knowledge	2
Q.4 Solve Any Two of the following.		12
A) Plan a dog legged stair for a building with the following data	Synthesis	
i) Vertical distance between the floors = 3.6 M		
ii) Size of stair hall = 2.5 M X 5 M		6
iii) Thickness of the floor slab = 140 MM		
iv) Thickness of the waist slab and landing slab = 100 MM		
B) What is the requirement of good staircase?	Understand	6
C) With neat sketch describe the technical terms used in door?	Application	6

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Q. 5	Solve of the following.		12
A)	State step by step procedure to construct concrete flooring for an industrial building.	Understand	6
B)	Write down the construction details for the tile flooring.	Understand	6
Q. 6	Solve of the following.		12
A)	What are the advantages and disadvantageous of precast structure?	Remember	6
B)	Explain the concept of erection in precast building.	Understand	6

***** End *****

**DR. BABASAHEB AMBEDKAR TECHNOLOGICAL
UNIVERSITY, LONERE - RAIGAD -402 103
Winter Semester Examination - Dec. - 2019**

Branch: Civil Engineering
Subject:- Building Construction (BTCVC305)
Date:- 19/12/2019

Sem.:- III
Marks: 60
Time:- 3 Hr.

Instructions to the Students

1. Each question carries 12 marks.
2. Attempt **any five** questions of the following.
3. Illustrate your answers with neat sketches, diagram etc., wherever necessary.
4. If some part or parameter is noticed to be missing, you may appropriately assume it and should mention it

(Marks)

- Q.1. a) Describe the characteristics of good stone masonry. (06)
- b) Differentiate between English Bond and Flemish Bond, and also draw neat sketches showing plan of consecutive courses 1 and 1/2 brick thick wall in single and double Flemish bond. (06)
- Q.2. a) Discuss the properties of fresh and hardened concrete. (06)
- b) Explain the purpose of admixture. State any four admixtures used in Concrete. (06)
- Q.3. a) Sketch and label the various components of an arch. (06)
- b) Explain the necessity of lintel along with its classification by material and Draw a labeled sketch of reinforced concrete lintel with chajja projection. (06)
- Q.4. a) Draw a neat sketch of fully paneled door in elevation for a opening size of 1200 mm x 2200 mm and enlist the various fixtures and fastenings for doors and windows. (06)
- b) Draw labeled sketch (sectional elevation) of dogged legged staircase for residential building. (06)
- Q.5. a) List and explain any four types of floor finishes with their suitability. (06)
- b) Explain the schematic process of erection work in high rise buildings. (06)
- Q.6. Solve any **four** of the following (4x3 M=12)
- a) Write a short note on the importance of partition walls with mentioning the types.
 - b) Explain the field tests for cement.
 - c) State the functions of window sill and lintel.
 - d) Write a short notes on ramps construction.
 - e) Suggest the roofing material for various types of pitched roofs with justification.
 - f) Write the advantages of pre-engineered buildings

Paper End

	DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Supplementary Examination – Summer 2022 Course: B. Tech. Branch : Semester :III Subject Code & Name: Building Construction (BTCVC305) Max Marks: 60 Date: Duration: 3 Hr.		
	Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.		
		(Level/CO)	Marks
Q. 1	Solve Any Two of the following.		
A)	Enlist various types of bonds in brick masonry. Explain with neat sketch any two bonds.	CO1	6
B)	Draw a neat sketch of uncoursed rubble masonry and explain it in brief.	CO1	6
C)	Explain the various chemical admixtures used in concrete.	CO2	6
Q.2	Solve Any Two of the following.		
A)	Explain the properties of i) Fresh concrete ii)Hardened concrete	CO2	6
B)	Write down the functions and requirements of good formwork.	CO3	6
C)	Draw a detailed labelled diagram of arch and explain components of arch.	CO3	6
Q. 3	Solve Any Two of the following.		
A)	Write short note on: i)Brick lintel ii)RCC lintel	CO3	6
B)	Enlist the types of doors. Explain any one type of door in detail with sketch.	CO3	6
C)	Explain with neat sketch Louvered window.	CO3	6
Q.4	Solve Any Two of the following.		
A)	Enlist the different types of stairs and explain any one with neat sketch.	CO3	6
B)	Describe the Escalator with neat sketch.	CO3	6
C)	State the various factors affecting the selection of flooring material.	CO3	6
Q. 5	Solve Any Two of the following.		
A)	Explain the Queen-post roof truss by sketch.	CO4	6
B)	State the various precautions and safety measures to be taken on	CO4	6

	construction site.		
C)	Explain the methods of Pre-fabrication.	CO4	6
	*** End ***		

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Supplementary Examination – Summer 2022

Course :B.Tech.

Branch : Civil Engg.

Semester : III

Subject Code & Name : BTCVC303 Building Construction and drawing

Max Marks : 60

Date : / /

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q.1 Solve Any Two of the Following		
A) Draw the labeled plan and cross section of uncoursed Rubble masonry?	CO1	06
B) Describe the following with sketches (a) Header stone (b) Queen closer (c) Mitred closer (d) Through stone	CO1	06
C) Explain various type of brick bond	CO1	06
Q.2 Solve Any Two of the Following		
A) What are the properties of hardened concrete?	CO2	06
B) Write a short note on cohesion and segregation.	CO2	06
C) Enlist various type of admixture and Explain any two admixture	CO2	06
Q.3 Solve Any Two of the Following		
A) Explain the types of Lintels and discuss about of two.	CO2	06
B) Draw the figure of arch & define any six technical terms in arch.	CO1	06
C) Describe primary and secondary and continuous beam.	CO2	06
Q.4 Solve Any two of the Following		
A) What is requirements of a good stair	CO4	06
B) Enlist various type of staircase and draw dog legged staircase plan and elevation with technical term.	CO4	06
C) What is type of elevators and their use?	CO2	06

Q.5 Solve Any Two of the Following

- | | | | |
|----|--|-----|----|
| A) | State step by step procedure to construct concrete flooring for an industrial building | CO2 | 06 |
| B) | Write down the construction details for the tile flooring. | CO2 | 06 |
| C) | What are the advantages and disadvantages of precast structure? | CO2 | 06 |

**DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY,
LONERE – RAIGAD -402 103
Semester Examination –2023**

Course: B. Tech in Civil Engineering
Subject Name: Building Construction & Drawing
Max Marks: 60 Date: 13/03/2023

Sem: III
Subject Code: BTCVC303
Duration: 3 Hr.

Instructions to the Students

1. Each question carries 12 marks.
2. Attempt **any five** question of the following.
3. Illustrate your answers with neat sketches, diagram etc., wherever necessary.
4. Assume suitable data wherever necessary and mention it clearly.
5. All sketches should be labeled, neat and proportionate

		(Marks)
Q.1.	Solve Any Two of the following.	
	a) What are the general principles observed in brick masonry?	(CO 1) 06
	b) State the comparative merits and demerits of stone masonry and brick masonry?	(CO 1) 06
	c) What are the general principles observed in Stone masonry?	(CO 1) 06
Q.2.	Solve Any Two of the following.	
	a) What are the ingredients of concrete? Enlist the properties of fresh & Hardened Concrete	(CO 2) 06
	b) Explain type of Admixtures & their necessity.	(CO 2) 06
	c) What are the chief ingredients of plain cement concrete? Describe in detail the properties of each?	(CO 2) 06
Q.3.	Solve Any One of the following.	
	a) Explain type formwork for RCC elements -Slab, Beam, Column,	(CO 3) 12
	b) Sketch a semicircular arch showing all technical terms used in arch? Sketch the entire arrangement showing the details of centering work?	(CO 3) 12
Q.4.	Solve Any Two of the following.	
	a) Draw a neat sketch of RCC stair showing all components?	(CO 4) 6
	b) Draw a neat sketch of Casement window?	(CO 4) 6
	c) Draw a neat sketch of Frame and Panelled door? (plan , section & elevation)	(CO 4) 6
Q.5.	Solve Any One of the following.	
	a) Explain king post truss and its joints with neat sketches?	(CO 5) 12
	b) Explain Queen post truss and its joints with neat sketches?	(CO 5) 12
Q.6.	Solve Any two of the following.	
	a) Explain merits and demerits of prefabrication?	(CO 5) 6
	b) Explain type of Flooring	(CO 5) 6
	c) Describe various guidelines for transportation and erection in prefabrication?	(CO 5) 6

*** End ***

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular and Supplementary Winter Examination– 2023

Course: B. Tech in Civil Engineering

Sem: III

Subject Name: Building Construction & Drawing

Subject Code: BTCVC303

Max Marks: 60

Date: 6/1/2024

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Solve Any Two of the following.		12
A) Describe general principles to be observed in stone masonry construction?	CO1	6
B) Draw a plan of 'L' shaped 1½ thk brick wall in English bond?	CO1	6
C) Explain any two types of partition walls with sketch?	CO1	6
Q.2 Solve Any Two of the following.		12
A) Draw a plan & section of a typical column with reinforcement?	CO2	6
B) Describe workability test of concrete with sketch?	CO2	6
C) Describe importance of chemical admixtures with an example?	CO2	6
Q. 3 Solve Any Two of the following.		12
A) Draw a sketch of semicircular arch & label all its parts?	CO2	6
B) Explain different types of formworks?	CO2	6
C) Explain any two types of lintels with sketch?	CO2	6
Q.4 Solve Any Two of the following.		12
A) Draw a sketch of louvered window?	CO3	6
B) Explain any three types of flooring?	CO3	6
C) Describe the role of Ramp with sketch?	CO3	6
Q. 5 Solve Any Two of the following.		12
A) Draw a sketch of king post truss?	CO4	6
B) What are the advantages and disadvantages of prefabricated structure?	CO4	6
C) Explain the term 'Tolerance' in prefabrication?	CO4	6

***** End *****